

Sahil Pawar

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EDUCATION

Trinity Academy of Engineering, SPPU

Bachelor of Engineering in Computer Science Engineering | CGPA: 8.51

Pune, India

Expected May 2027

EXPERIENCE

Innomatics Research Labs

Remote

Agentic AI Intern

Feb 2026 – May 2026

- Developed a FastAPI backend for a resource management system, using SQLAlchemy row-level locking to prevent race conditions when multiple requests modified the same membership record simultaneously.
- Built a local RAG pipeline with LangGraph and ChromaDB, chunking documents recursively and retrieving the top-k most relevant matches via cosine similarity to ground LLM responses in real context.
- Implemented a human-in-the-loop fail-safe that routes AI outputs below a 0.5 confidence threshold to a structured review queue instead of surfacing an uncertain answer directly to users.

PROJECTS

HealthScribe | Django, FastAPI, React, LangChain, ChromaDB, Gemini 2.5 Flash, Sarvam AI, Supabase

Demo

- Built a full-stack medical records platform extracting and verifying prescriptions via a hybrid OCR pipeline (Sarovam AI + Gemini 2.5 Flash), and a LangChain + ChromaDB RAG chatbot for querying patient history.
- Benchmarked hybrid OCR-LLM extraction against a Gemini-only baseline across 4 clinical entity categories; achieved **94%** exact-match accuracy on medicines vs. an **85%** baseline, at a **1.46x** latency cost (**29.2s** vs. **20.0s**).
- Replaced similarity search with MMR retrieval in ChromaDB, doubling clinical category coverage and cutting duplicate-record redundancy from **62%** down to **8%**; validated results using a Gemini-as-a-Judge evaluation framework.
- Implemented allergy-drug conflict alerts, LLM-based drug interaction checking, multi-doctor treatment comparison, and a shareable QR-based health profile.

VeriFrame | FastAPI, LangGraph, HuggingFace Transformers, OpenCV, MediaPipe, Groq API, React, Supabase

Demo

- Built a multi-agent video forensics platform detecting AI-generated video and deepfakes using a Vision Transformer, OpenCV/MediaPipe temporal analysis, and Groq Llama 4 Scout reasoning orchestrated with LangGraph – producing explainable, per-frame verdicts instead of a single score.
- Evaluated on a benchmark of 40 real and face-swapped videos, achieving **87.5%** overall accuracy and a **90.0%** deepfake detection rate, with a **15.0%** false positive rate.
- Reduced end-to-end latency to **3.63s per video** by downscaling uploads to **480p** and extracting frames at **0.5s** intervals, then routing only the top **8** highest-risk frames to the LLM, keeping API cost proportional to anomaly count rather than video length.
- Designed dynamic weight redistribution in the consensus engine: if the local ViT model fails to load, its weight drops from **40%** to **10%** while the LLM's rises to **60%**, degrading to a still-functional LLM-primary mode instead of returning unreliable results.

ACHIEVEMENTS

- Achieved **Knight status** on LeetCode and solved **500+** Data Structures and Algorithms problems across various coding platforms. (Profile)
- Event Coordinator, Association of Computer Engineering Students (ACES)** – Organized department-wide coding contests for 100+ students and coordinated technical workshops.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

Frameworks & Libraries: Django, FastAPI, React, DRF, LangChain, LangGraph, SQLAlchemy, OpenCV, MediaPipe, Pydantic

AI/GenAI: Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Cross-Encoder Reranking, LLM-as-a-Judge Evaluation, HuggingFace Transformers, Structured JSON Outputs, Gemini API

Databases: PostgreSQL, Supabase, ChromaDB

Tools & Technologies: Git, GitHub, LangSmith, Postman, JWT, REST APIs, pytest